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(54) **ROTATING BLADE FOR A TURBOMACHINE**

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(71) Applicant: **MTU Aero Engines AG**, Munich (DE)

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(72) Inventors: **Manfred Dopfer**, Unterschleissheim (DE); **Salome Gassmann**, Munich (DE); **Johannes Linhard**, Andechs (DE)

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(73) Assignee: **MTU Aero Engines AG**, Munich (DE)

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Primary Examiner — Sang K Kim
(74) *Attorney, Agent, or Firm* — Hinckley, Allen & Snyder, LLP; David R. Josephs

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(57) **ABSTRACT**

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F04D 29/32 (2006.01)

The present invention relates to a rotating blade for a turbomachine, having a blade element and a shroud, wherein the shroud is profiled at a peripheral side, i.e., when observed in a tangential section, has a contact flank that is oriented axially, at least proportionally, and has a free flank that is also oriented axially, at least proportionally, and is in fact opposite the contact flank, wherein, when observed in the tangential section, an intermediate segment between the contact flank and the free flank has a first curvature in a first transition segment to the contact flank and has a second curvature in a second transition segment to the free flank, wherein the first curvature is greater than the second curvature.

(52) **U.S. Cl.**

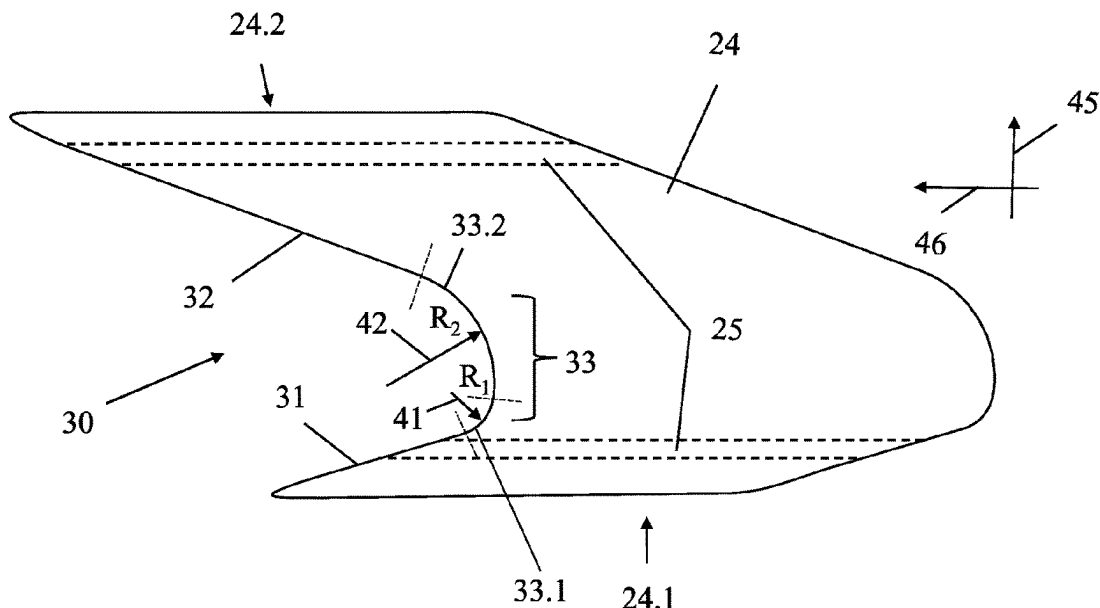
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CPC F01D 5/225; F05D 2240/11; F04D 19/02; F04D 29/324

See application file for complete search history.

16 Claims, 4 Drawing Sheets



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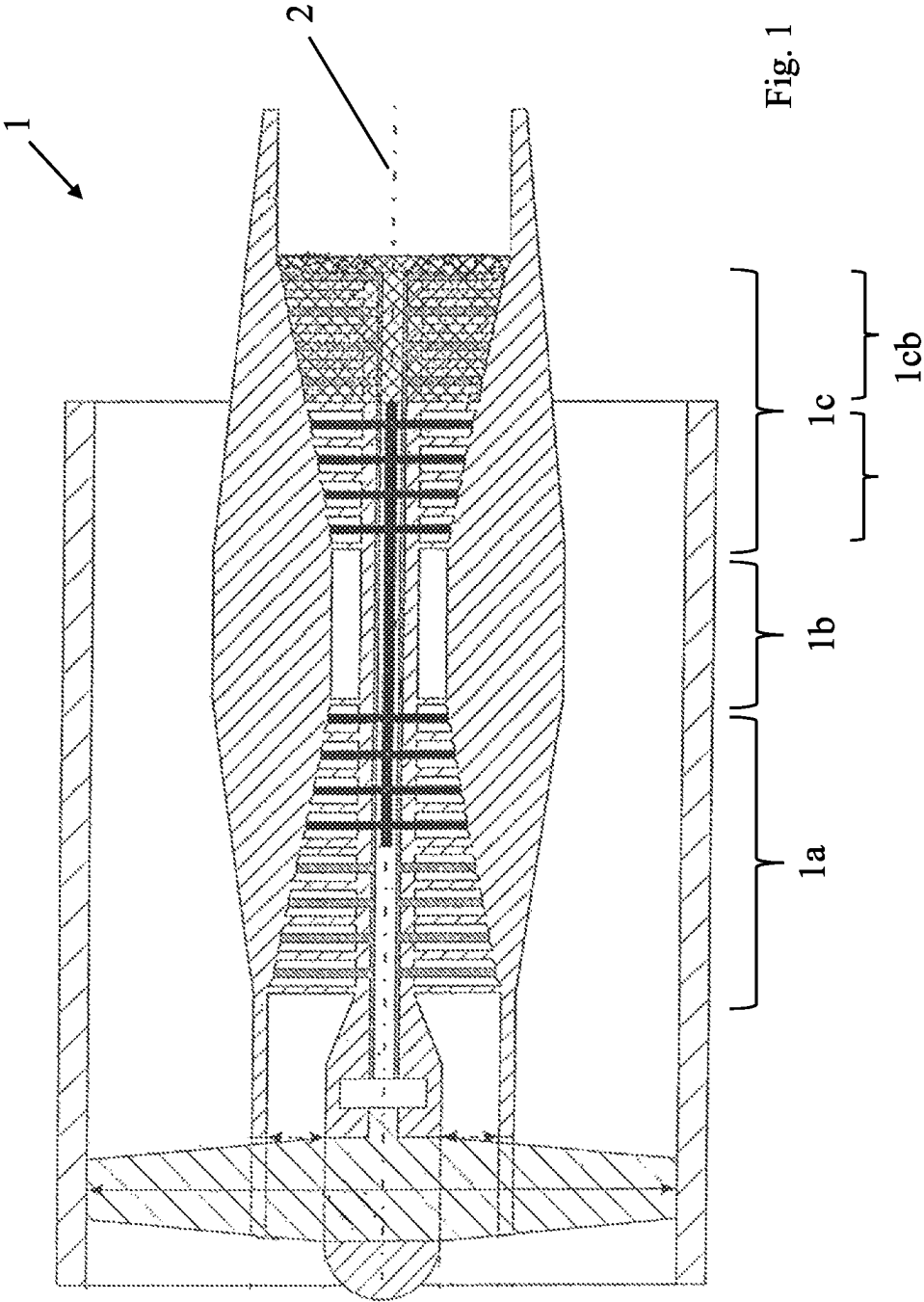


Fig. 1

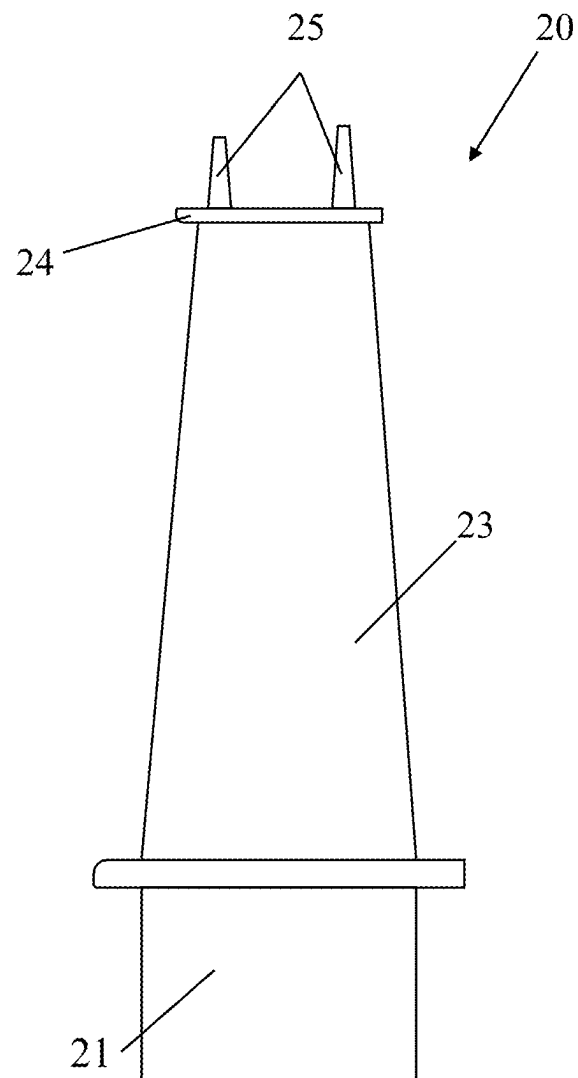


Fig. 2

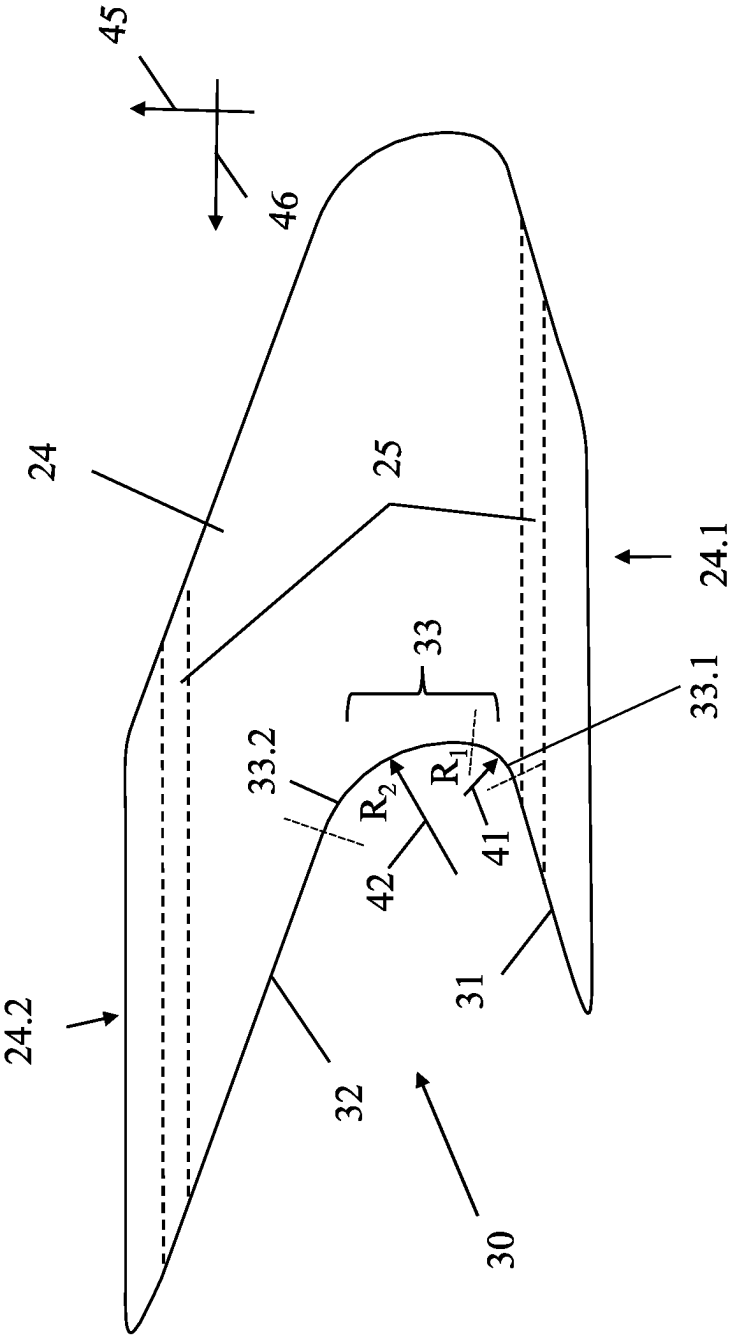


Fig. 3

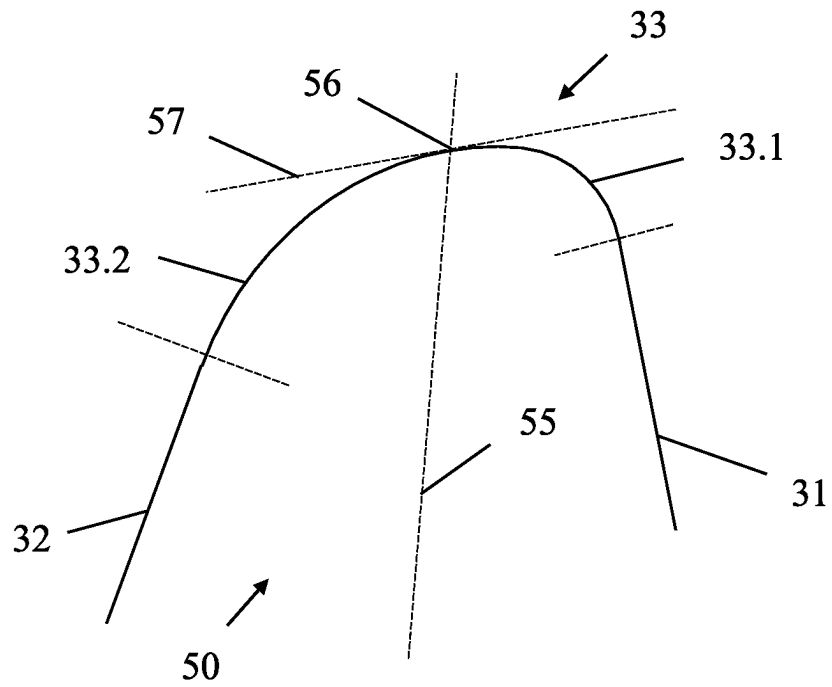


Fig. 4

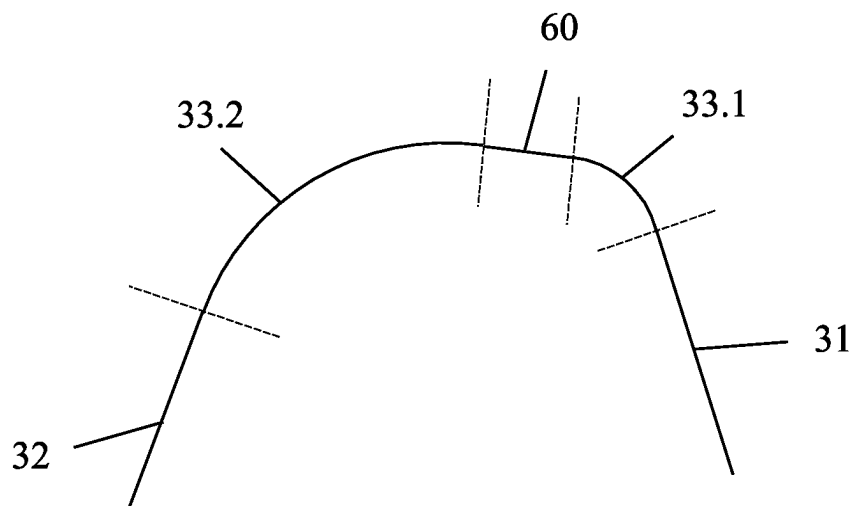


Fig. 5

ROTATING BLADE FOR A TURBOMACHINE

BACKGROUND OF THE INVENTION

The present invention relates to a rotating blade for a turbomachine.

The turbomachine may involve, for example, a jet engine, e.g., a turbofan engine. Functionally, turbomachines are divided into compressor, combustion chamber, and turbine. That is, in the case of the jet engine, intake air is compressed by the compressor and burned in the downstream combustion chamber with kerosene mixed in. The hot gas that arises, a mixture of combustion gas and air, flows through the downstream turbine and is thus expanded. Usually, the turbine is constructed of several stages, each with a stator (guide vane ring) and a rotor (rotating blade ring); the rotors are propelled by the hot gas. Here, in each stage, internal energy is withdrawn proportionally from the hot gas, this energy being converted into movement of the respective rotating blade ring and thus of the shaft.

The present subject relates to a rotating blade for arrangement in the gas channel of the turbomachine. The rotating blade can generally also find use in the compressor region and therefore it can be arranged in the compressor gas channel; if a use is preferred in the turbine region, it is thus placed in the hot-gas channel.

SUMMARY OF THE INVENTION

The present invention is based on the technical problem of providing a particularly advantageous rotating blade.

According to the invention, this problem is solved by the rotating blade and concretely by the structuring of the shroud as discussed below. The latter is profiled on a peripheral side, i.e., found there is an at least proportionally axially oriented, particularly straight or linear free flank, and lying opposite to this, a particularly straight or linear contact flank. In this case, an intermediate segment, which marks the transition between free and contact flanks, when viewed in a tangential section, is made up of different curvatures, that is, it is curved more in a first transition segment to the contact flank than in a second transition segment to the free flank.

This course of curvature or the combination of less curvature in the transition to the free flank and greater curvature in the transition to the contact flank can be of advantage with respect to the mechanical stress level in the shroud, and at the same time, assure a sufficient attachment surface or contact surface to the adjacent shroud. For illustration, if one considers the mechanical stresses alone, in the intermediate segment, a curvature that is as small as possible would be of interest, thus, for example, a radius of curvature that is as large as possible. Therefore, however, if a distance that the groove base lying in the intermediate segment has from a point of intersection of the flanks should increase arbitrarily, in other words, the intermediate segment would thus increasingly be displaced into the flanks.

Therefore, in particular, if the length of the contact flank should also decrease, the attachment surface to the adjacent shroud should thus decrease, and correspondingly, a mechanical stress resulting from the input of force as a consequence of the attachment should increase. And vice versa, with the present subject, thus of the proportionally greater curvature in the transition to the contact flank, whose length can be maintained or at least not excessively reduced, even though the transition to the free flank is made with a smaller curvature, thus for example, a larger radius. Fig-

ratively speaking, more material remains in the base of the groove based on this curvature combination.

Preferred embodiments are discussed below and the entire disclosure, wherein, in the presentation of the features, a distinction is not always made individually between device, method, and/or use aspects; in any case, the disclosure is to be read implicitly with respect to all claim categories. For example, if the advantages of the rotating blade are described in a specific application, this is to be read as a disclosure both of the correspondingly designed rotating blade and use of such.

The terms “axially”, “radially” and “circumference” or “circumferentially”, as well as the directions belonging thereto (axial direction, etc.) refer to the axis of rotation around which the rotating blade rotates during operation, unless it is expressly indicated to the contrary. The latter typically coincides with the longitudinal axis of the turbomachine, in particular of the engine or the engine module. The “tangential section” is taken in a sectional plane that is tangential to the circumference and here is parallel to the axis, this plane running through the shroud and, in particular, the peripheral side thereof. Preferably, the peripheral side is profiled correspondingly over its entire radial height; thus, the criterion according to the present invention is fulfilled in all tangential sections running through the peripheral side.

The “peripheral side” of the shroud points in the circumferential direction; by its peripheral side, the shroud borders the adjacent shroud (the adjacent rotating blade). The contact flank, for example, can be arranged closer to the axially front side of the shroud (and vice versa, the free flank can be arranged closer to the axially back side); referred to the through flow of the gas channel, the contact flank can thus lie upstream of the free flank (and vice versa, the free flank can lie downstream of the contact flank). The free flank is “axially oriented at least proportionally; a normal line to the surface of the free flank thus has at least one directional component in the axial direction, particularly axially to the front.

Preferably, the free flank lies diagonally (not perpendicularly) to the axial direction, and the surface normal line can enclose with the latter, for example, an angle of at least 20° or 25°, whereby possible upper limits can lie, for example, at most at 45°, 40° or 35° (considered here each time as the smaller of two angles enclosed between the normal line and the axial direction. The contact flank is also axially oriented at least proportionally; its surface normal line therefore also has at least one directional component in the axial direction. Since it is arranged lying opposite axially to the free flank, the surface normal line points in the axially opposite direction, particularly to the back axially. The contact flank preferably also lies diagonally (not perpendicularly) to the axial direction; it can enclose with the free flank, for example, a cutting angle of at least 55° or 60°, with possible upper limits, for example, of at most 75° or 70°.

Generally, in the scope of this disclosure, “a” and “an”, unless expressly indicated otherwise, are to be read as the indefinite article and thus also are always to be read as “at least one”; thus, for example, at least two blade elements also can be provided at the shroud (“multi-segment”). In general, the profiled shroud according to the invention can also be a radially inner shroud arranged at the blade element, but preferably it involves an outer shroud (radially outer at the blade element). A sealing tip, that is, a sealing ridge extending in the circumferential direction, i.e., a sealing tip axially in front and another axially in back, can be provided radially outside at the shroud.

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According to a preferred embodiment, the intermediate segment has a central segment that lies between the more greatly curved first transition segment and the less greatly curved second transition segment and extends linearly when viewed in the tangential section. Preferably, the central segment transitions tangentially into the first and/or second transition segment; thus, in other words, the first and the second curvatures can be joined together via a tangent. For example, the central segment can create a freedom of design or customization option, i.e., in the integration of the differently pronounced curvatures into an existing design.

In an alternatively preferred embodiment, the first and the second transition segments transition directly into one another, therefore without a straight piece in between them. The transition is preferably tangential in this case; the first and second transition segments then have a common tangent at the transition point.

In a preferred embodiment, the first transition segment has a constant first radius of curvature and/or the second transition segment has a constant second radius of curvature. Preferably, both apply, wherein the first radius of curvature is then less than the second, and correspondingly, when the first curvature is greater than the second. Alternatively to a constant radius or radii, in contrast, generally any course is conceivable, for example, an elliptical shape, or even free form, or fitted with splines. In comparison to this, "ruled geometry", thus shaping with circles or radii and optionally a straight line, can be easier to be measured, for example; also drawings can be more easily prepared or examined, in any case with standard procedures.

According to a preferred embodiment, the intermediate segment is made up exclusively by the first and the second radii of curvature; therefore, there are no other curvatures. Optionally, the linear or tangent piece can be provided therebetween; see above.

According to an embodiment, when observed in a tangential section from the particularly linear contact flank to the particularly linear free flank, every curvature of the peripheral side has the same sign; it is concave in particular, and/or, when observed in a tangential section from the particularly linear contact flank to the particularly linear free flank, every transition from a curved to a linear section is tangential.

Preferably, when observed in a tangential section, the change of course of the peripheral side from the particularly linear contact flank to the particularly linear free flank is always differentiable; thus it has a kink, an abrupt change, or a discontinuity.

In a preferred embodiment, the first transition segment transitions tangentially into the particularly linear contact flank, and/or the second transition segment transitions tangentially into the particularly linear free flank. Observed in the tangential section, a linear extension of the free flank can thus be a tangent at a circle placed in the second transition segment (with a second radius) and/or a linear extension of the contact flank can be a tangent at a circle placed in the first transition segment (with a first radius). The tangential transition can be of advantage, for example, with respect to mechanical stresses.

In a preferred embodiment, a groove formed with the intermediate segment is effectively asymmetric, so that an angle bisecting line lies diagonally to a tangent at the base of the groove. The angle bisecting line lies in the middle between the contact flank and the free flank, and it bisects the intermediate segment by definition in the base of the groove. If one applies a tangent at this point of intersection

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to the intermediate segment, the latter lies diagonally and thus not at a right angle to the angle bisecting line.

In a preferred embodiment, when viewed optionally from the linear or tangent piece, the intermediate segment is at the most concavely curved; therefore, there is no convexly curved segment. In particular, in connection with the one or more tangential transitions of the transition segments into the contact flank and the free flank, in the "contact flank-intermediate segment-free flank" region, the peripheral side can be provided without a convex curvature overall, which can offer advantages, for example, with respect to leakage. Based on the shaping without curvature reversal or alternation or discontinuity, e.g., a complementary shaping of the adjacent shroud can be simplified and correspondingly, a tighter attachment can be achieved.

In general, the opposite peripheral side of the shroud, which lies opposite to the profiled peripheral side in the circumferential direction, is preferably profiled in a complementary manner. In a rotating blade ring, the rotating blade concerned can be assembled circumferentially with additional rotating blades or segments, wherein, preferably, shrouds of the same construction are placed next to each other circumferentially.

In addition to a rotating blade ring, the subject also relates to a turbine module, in particular to a turbine module for an aircraft engine. For example, the latter may involve a geared turbofan engine, based on coupling by way of the gear; in particular, during operation, the turbine module can rotate faster than the fan of the aircraft engine (it can thus be "rapid-running"). This may involve the turbine module, for example, a high-pressure or, in particular, a low-pressure turbine module.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will be explained in more detail in the following on the basis of an example of embodiment, wherein the individual features in the scope of the present invention can also be essential to the invention in another combination, and wherein also no distinction is made individually between the different claim categories. Taken individually,

FIG. 1 shows schematically a turbofan engine in an axial section;

FIG. 2 shows a rotating blade with blade element and shroud in a schematic lateral view;

FIG. 3 shows the shroud of the rotating blade according to FIG. 2 in a tangential section;

FIG. 4 shows a view of a detail of FIG. 3;

FIG. 5 shows an alternative embodiment of the profiling of FIG. 4.

DESCRIPTION OF THE INVENTION

FIG. 1 shows a turbomachine 1 in a schematic view, concretely a turbofan engine. The turbomachine 1 is functionally divided into compressor 1a, combustion chamber 1b and turbine 1c; the latter comprises a high-pressure turbine module 1ca and a low-pressure turbine module 1cb. In this case, both the compressor 1a as well as the turbine 1c are each constructed of several stages; each stage is composed of a guide vane ring and a rotating blade ring. During operation, the rotating blades rotate around the longitudinal axis 2.

FIG. 2, in a lateral view, shows a rotating blade 20 of the turbine 1c, concretely of the low-pressure turbine module 1cb. The latter comprises a rotating blade element 23 and a

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shroud **24**; it is mounted on a disk (not shown here) by way of a blade root **21**. Sealing tips **25**, which, for example, run into a run-in lining of the housing (not shown) during operation, are arranged radially outside on the shroud **24**.

FIG. **3** shows the shroud **24** in a tangential section, wherein its axial front end **24.1** lies on the left in FIG. **2** and the axial back end **24.2** lies on the right in FIG. **2**. Further, for orientation, in FIG. **3**, the sealing tips **25** are shown, but they lie outside the plane of section and thus are reproduced by dotted lines. The subject is particularly directed to the profiling of the peripheral side **30** of the shroud **24**. The latter is assembled with a free flank **32** and a contact flank **31** that lies opposite axially, as well as an intermediate segment **33** formed therebetween.

The intermediate segment **33** is composed of a first transition segment **33.1** that runs tangentially into the contact flank **31** and has a first curvature **41**, which is greater than a second curvature **42**, by which a second transition segment **33.2** runs into the free flank **32**. The curvatures **41**, **42** are realized here by way of a first and a second radius of curvature R_1 , R_2 , wherein the greater curvature is associated with a smaller radius. The combination of smaller and greater curvature can be of advantage, on the one hand, with respect to the mechanical stress level in the shroud **24**, wherein, on the other hand, the extent of the contact flank **31** and thus attachment to the adjacent shroud is at least not essentially reduced; see the introduction to the description in detail. For orientation in FIG. **3**, an axial direction **45** as well as a circumferential direction **46** are also shown.

FIG. **4** shows the groove **50** formed between the contact flank **31** and the free flank **32** once more in a detailed view (also in a tangential section). Shown further is an angle bisecting line **55** that bisects the intermediate segment **33** in the base **56** of the groove, a line that lies in the middle between contact flank **31** and free flank **32**. A tangent **57** that is indicated there lies diagonally due to the asymmetric shaping, thus not at a right angle to the angle bisecting line.

FIG. **5** shows a slightly modified embodiment, in which the first and the second transition segments **33.1**, **33.2** do not directly transition into one another. Instead of this, a central segment **60** is formed therebetween, which extends linearly and in each case tangentially into the two transition segments **33.1**, **33.2**.

What is claimed is:

1. A rotating blade for a turbomachine, comprising a blade element and a shroud, wherein the shroud is profiled at a peripheral side when observed in a tangential section,
 - has a linear contact flank that is oriented axially, at least proportionally, and
 - has a linear free flank that is also oriented axially, at least proportionally, and is opposite the contact flank,
 wherein, when observed in the tangential section, an intermediate segment between the contact flank and the free flank in a first transition segment to the contact flank has a first curvature, the contact flank being tangential to the first curvature where the contact flank meets the first transition segment, and

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in a second transition segment to the free flank has a second curvature, wherein the first curvature is greater than the second curvature, and wherein each curvature between the contact flank and the free flank is concave.

2. The rotating blade according to claim 1, in which the intermediate segment between the first and the second transition segments has a central segment, which runs linearly when observed in the tangential section.

3. The rotating blade according to claim 1, wherein, in the intermediate segment, the first and the second transition segments transition directly into one another.

4. The rotating blade according to claim 1, wherein the intermediate segment, when observed in the tangential section, has a constant first radius of curvature R_1 over the first transition segment.

5. The rotating blade according to claim 4, wherein the intermediate segment, when observed in the tangential section, apart from a linear extension that is in sections, is included exclusively with the first and the second radii of curvature R_1 , R_2 .

6. The rotating blade according to claim 1, wherein the intermediate segment, when observed in the tangential section, has a constant second radius of curvature R_2 over the second transition segment.

7. The rotating blade according to claim 6, wherein the intermediate segment, when observed in the tangential section, apart from a linear extension that is in sections, is included exclusively with the first and the second radii of curvature R_1 , R_2 .

8. The rotating blade according to claim 1, wherein, when observed in a tangential section, from the contact flank to the free flank, every transition from a curved to a linear segment is tangential.

9. The rotating blade according to claim 1, wherein the first transition segment, when observed in the tangential section, transitions tangentially into the contact flank.

10. The rotating blade according to claim 1, wherein the second transition segment, when observed in the tangential section, transitions tangentially into the free flank.

11. The rotating blade according to claim 1, wherein, when observed in the tangential section, an angle bisecting line that lies in the middle between the contact flank and the free flank and bisects the intermediate segment in a groove base, lies diagonally to a tangent placed in the groove base of the intermediate segment.

12. The rotating blade according to claim 1, wherein the intermediate segment, when observed in the tangential section, apart from a linear extension optionally in sections, is curved exclusively concavely.

13. A rotating blade ring having a rotating blade according to claim 1.

14. A turbine module having a rotating blade ring according to claim 13.

15. A turbine module having a rotating blade according to claim 1.

16. A jet engine, having a turbine module according to claim 15.

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